

Rectified Flow for Chromosome Detection: Stable Coupling and Few-Step Inference

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Abstract—Chromosome karyotyping is the first-line diagnostic test for hematological malignancies and congenital syndromes, yet manual analysis requires hours per specimen. Diffusion-based object detectors offer a probabilistic framework for chromosome detection but suffer training instability caused by optimal-transport (OT) coupling diversity collapse. We propose a Rectified Flow (RF) detector with Stochastic Coupling, theoretically grounded by a diversity-collapse analysis, and accelerated by a DPM-Solver++ two-step sampler with Top-K box pruning. On two public chromosome datasets, our method improves mAP by 0.082 over DiffusionDet while reducing inference to 2 steps at X FPS. The stabilizer eliminates seed-level variance, enabling clinically deployable, real-time metaphase analysis. Code and trained models are released to support reproducibility.

Index Terms—Chromosome detection, rectified flow, diffusion models, optimal transport, few-step inference, karyotyping, medical image analysis.

I. INTRODUCTION

CHROMOSOME analysis via karyotyping is the first-line test for hematological malignancies and congenital syndromes [?]. Manual karyotyping takes a skilled cytogeneticist several hours per specimen, creating a throughput bottleneck in clinical genetics laboratories. Recent object-detection frameworks based on denoising diffusion models (e.g., DiffusionDet [?]) offer a unified probabilistic formulation for detection, but their training is sensitive to coupling diversity and convergence is brittle on dense, small-object modalities such as metaphase chromosomes.

II. RELATED WORK

III. METHODOLOGY

- A. Rectified Flow for Object Detection
- B. Stochastic Coupling: Resolving OT Diversity Collapse
- C. DPM-Solver++ 2-Step Inference with Top-K Pruning

IV. EXPERIMENTS

- A. Datasets
- B. Evaluation Metrics
- C. Comparison with State-of-the-Art
- D. Ablation Studies

V. DISCUSSION

VI. CONCLUSION

ACKNOWLEDGMENTS

REFERENCES

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